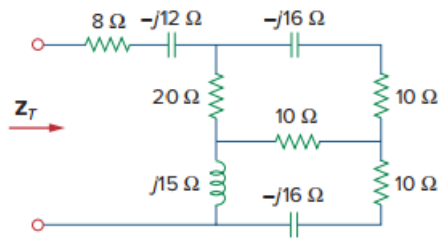


Problem

For the circuit in Fig. 9.70, find the value of Z_T .

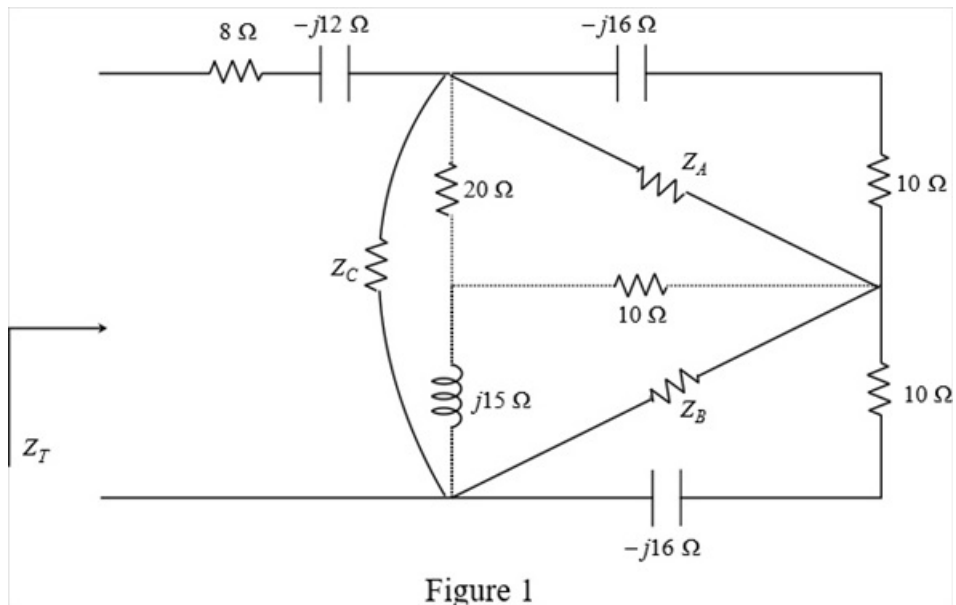


Step-by-step solution

Step 1 of 5

Refer to Figure 9.70 in the textbook.

Replace the WYE network composed of the $20\ \Omega$, $10\ \Omega$, $j15\ \Omega$ impedances into its corresponding delta network as shown in Figure 1.



Step 2 of 5

Calculate Z_A , Z_B and Z_C .

$$\begin{aligned}Z_A &= \frac{(20)(10) + (10)(15j) + 30(10j)}{j15} \\&= \frac{200 + 450j}{15j} \\&= 30 - j13.33 \, \Omega \\Z_B &= \frac{(20)(10) + (10)(15j) + 30(10j)}{20} \\&= \frac{200 + 450j}{20} \\&= 10 + j22.5 \, \Omega \\Z_C &= \frac{(20)(10) + (10)(15j) + 30(10j)}{10} \\&= \frac{200 + 450j}{10} \\&= 20 + 45j \, \Omega\end{aligned}$$

Step 3 of 5

Z_A is in parallel combination with $(10 - j16) \, \Omega$ impedance.

$$Z_{eq1} = Z_A \parallel (10 - j16)$$

Substitute $30 - j13.33 \, \Omega$ for Z_A .

$$\begin{aligned}Z_{eq1} &= (30 - j13.33) \parallel (10 - j16) \\&= \frac{(30 - j13.33)(10 - j16)}{(30 - j13.33) + (10 - j16)} \\&= 8.721 - j8.938 \, \Omega\end{aligned}$$

Z_B is in parallel combination with $(10 - j16) \, \Omega$ impedance.

$$Z_{eq2} = Z_B \parallel (10 - j16)$$

Substitute $10 + j22.5 \, \Omega$ for Z_B .

$$\begin{aligned}Z_{eq2} &= (10 + j22.5) \parallel (10 - j16) \\&= \frac{(10 + j22.5)(10 - j16)}{(10 + j22.5) + (10 - j16)} \\&= 21.70 - j3.821 \, \Omega\end{aligned}$$

Step 4 of 5

Draw the reduced equivalent circuit as shown in Figure 2.

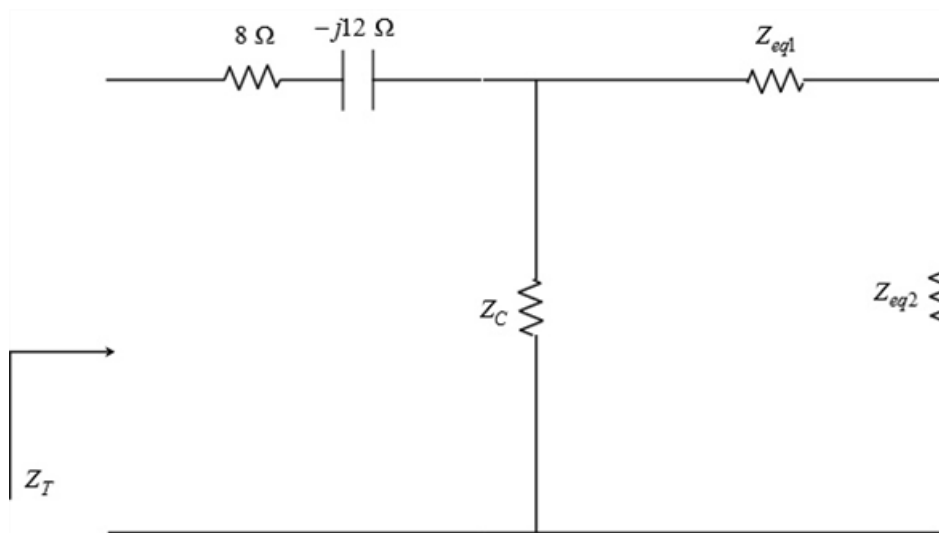


Figure 2

Step 5 of 5

Z_C is in parallel combination with $(Z_{eq1} + Z_{eq2})$.

$$Z_{eq3} = Z_C \parallel (Z_{eq1} + Z_{eq2})$$

Substitute $20 + 45j$ for Z_C , $8.721 - j8.938$ for Z_{eq1} and $21.70 - j3.821$ for Z_{eq2} .

$$\begin{aligned} Z_{eq3} &= (20 + 45j) \parallel (8.721 - j8.938 + 21.70 - j3.821) \\ &= \frac{(20 + 45j)(30.421 - 12.759j)}{20 + 45j + 30.421 - 12.759j} \\ &= 26.38 + j4.84 \Omega \end{aligned}$$

Z_T is the series combination of $8 - 12j$ and Z_{eq3} .

Calculate Z_T .

$$Z_T = 8 - 12j + Z_{eq3}$$

Substitute $26.38 + j4.84 \Omega$ for Z_{eq3} .

$$\begin{aligned} Z_T &= 8 - 12j + 26.38 + j4.84 \\ &= 34.38 - j7.16 \Omega \end{aligned}$$

Hence, the value of Z_T is $\boxed{34.38 - j7.16 \Omega}$.